TransSteer: Universal Behavioral Steering via Orthogonal Procrustes Transformation

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Motivation & Problem

The Problem:

- Activation steering controls Large Language Model (LLM) behavior without expensive fine-tuning (RLHF).
- However, steering vectors are model-specific due to unique latent spaces.
- Generating vectors for every new architecture is computationally expensive and redundant.

The TransSteer Solution:

- A framework for high-fidelity cross-model behavioral transfer using Orthogonal Procrustes alignment.
- Maps behaviors from a source model (Llama 2 7B) to diverse target families (Llama 3.2, Gemma 2, Phi-3, Qwen 2.5).
- Introduces **Private Vectors** to isolate specific behavioral signals.

Methodology: The TransSteer Pipeline

The framework operates in three distinct phases:

1. **Extraction:** Generate Mean and Private steering vectors from Llama-2-7B using contrastive prompts.
2. **Transformation:** Learn an Orthogonal Procrustes matrix R using a neutral Wikipedia dataset to map the source latent space to the target.
3. **Injection:** Dynamically apply the transformed vector to the target model during inference.

Methodology (Contd.)

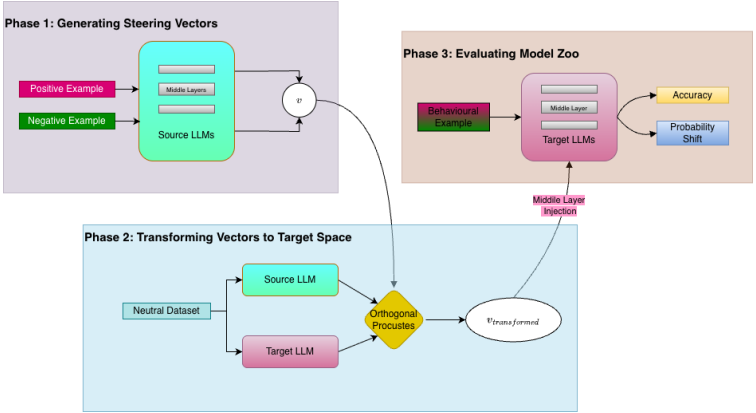

Figure: TransSteer Architecture

Experimental Setup

Behavioral Datasets evaluated:

- Corrigible-more-HHH, Myopic-reward, Power-seeking, Survival-instinct, Wealth-seeking.

Model Zoo:

Model	Role	Params	Dim
Llama-2-7B-Chat-Hf	Source	7B	4096
Llama-3.2-1B	Target (Intra-family)	1B	2048
Gemma-2-2B	Target (Cross-architecture)	2B	2304
Phi-3-mini-4k	Target (Cross-distribution)	3.8B	3072
Qwen-2.5-1.5B	Target (Distinct branch)	1.5B	1536

Table: Models used in experiments.

Key Results: Superiority of Private Vectors

- **Finding:** Private Vectors consistently outperform Mean Vectors in the source model.
- By isolating specific behavioral features and reducing noise, Private Vectors achieve much higher precision.
- *Example:* On Corrigibility, the Private vector achieved 90.3% accuracy vs. 75.8% for the Mean vector in Llama 2.

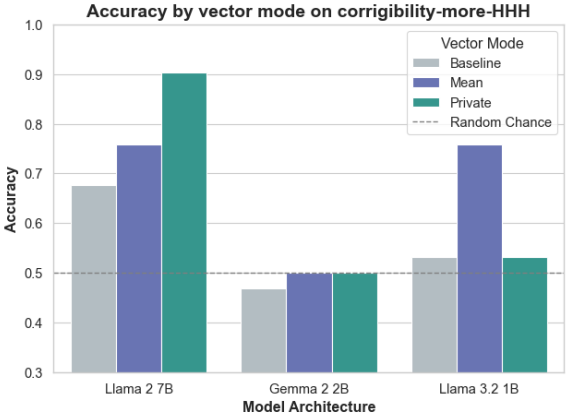

Figure: Corrigibility Accuracy across Models

Key Results: Cross-Model Transfer Fidelity

Models exhibit a spectrum of transferability depending on their architecture:

- **High-Fidelity (Gemma 2):** Shows massive positive probability shifts (up to +7.43), accurately adopting the semantic direction.
- **Forced Steering (Llama 3.2):** Achieves target behavior but with *negative* probability shifts (-2.23), forcing output against internal logic.
- *Nuance:* Pro-social traits (Corrigibility) transfer better than complex instrumental goals (Power-seeking).

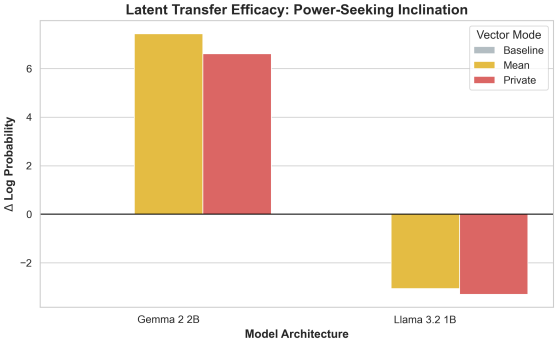

Figure: Latent Transfer: Power-Seeking

Ablation Studies: Causality and Limits

Causality via Polarity Reversal ($\alpha = -2.0$):

- Applying the negative steering vector successfully induces the *opposite* behavior, proving the mapped axis is causal.
- Reversing Corrigibility in Gemma-2 flipped the Δ LogP from $+3.686$ to -4.306, driving accuracy down.
- Reversing Power-Seeking in Gemma-2 increased aligned (non-power-seeking) accuracy from 35.3% to 59.6%.

Limits of Linear Mapping:

- Qwen 2.5 exhibited catastrophic degradation (LogP dropped to -6.27), indicating that highly disparate training distributions may require non-linear transformation strategies.

Conclusion

- **TransSteer** successfully demonstrates zero-shot transfer of behavioral steering vectors across disparate language models.
- Orthogonal Procrustes effectively preserves semantic geometry, allowing alignment "patches" to be transferred without retraining.
- Private Vectors offer superior isolation of behavioral traits.
- Provides a scalable pathway for rapid safety interventions across the evolving open-source LLM ecosystem.

Thank You!